

Stats	When it appears	Value
		C = same as in "Today" D = daily goal
List progress	When the user is learning a list (not History or Favorites)	E/F learned G mastered E = nb terms seen from list, excluding ignored terms (NbView ≠ 0 and isIgnored = False) F = all terms from the list, excluding ignored terms (All list terms - isIgnored = True) G = all mastered terms from the list, excluding ignored terms (isMemorized = True and isIgnored = False) From the endpoint : https://context.reverso.net/bst-web-user/user/favourites?listId=326721 E = learningStatus : inProgressCards F = learningStatus : totalCards G = memorizedOnceCards

Being able to fill the progress bar fully

Games issues

We have too many flashcard components and logics to update Learn DB. The goal of this section is to have one clear, consistent behavior.

What is Spaced Repetition

Have you ever crammed for a test, remembered everything for a few hours... and then forgotten it all the next day? That's because your brain needs time and repeated exposure to truly lock in new information. This is where **spaced repetition** comes in. It's a science-backed learning technique that helps you retain information for the **long term** — not just for a few hours.

Instead of reviewing a word over and over in one session, spaced repetition spreads your reviews out over **increasing intervals**. Each time you successfully remember the word after a break, your brain strengthens the memory even more.

The more you review a word at just the right time, the more deeply it sticks — until it pops into your mind effortlessly.

Here's an example:

Review	Time After Learning
1st review	1 day later
2nd review	3 days later
3rd review	7 days later
4th review	14 days later
5th review	Memorized

SRS Manager API

This API should do 3 things:

1. Fill the Learn DB with enough terms to play a game (if it's not already the case)
2. Suggest 7 terms to learn in a game
3. Update the Learn DB for each of the 7 terms based on the user's answers (ex. correct / false)

→ API input

The API takes as input 3 parameters:

1. Direction
ex. en → fr, en → en, ar

⇨ API output

1. The API will take the best X terms to learn based on chosen strategy

→ es

2. List to study

Possible values: All,
History, Favorites,
Particular list

3. Learning strategy

Possible values: New,
Mixed, SRS

4. Nb of terms to review

ex. 7, 10, 15...

2. The API will update the
Learn DB terms based
on user's actions and
answers

Algorithm for terms selection

How are terms selected?

Based on the chosen strategy, we select the terms:

Learning strategy	Description	Terms selection
SRS	<i>Reinforce what you've learned before moving on to new cards with spaced repetition.</i>	Prioritize terms to review based on the SRS priorities table below: Insert mistakes first Fill remaining slots with: • 25% New terms (by most recent) • 50% Fresh review terms (pas due ≤ 30d) • 25% Old review terms (past due > 30d)
NEW	<i>Study cards you haven't learned yet, starting with the most recently added.</i>	100% new terms • On web, prioritize by most recent • On mobile app, prioritize by most recent by default, but offer user a choice to have start from oldest or random IF not enough terms to review, switch

Learning strategy	Description	Terms selection
		to SRS • IF you have ≥ 3 new terms, you play a mini game • IF you have < 3 new terms, we switch to SRS
MIXED (default)	<i>A mix of new cards and cards to review.</i>	Insert mistakes first Fill remaining slots with: • 40% New terms (by most recent) • 50% Fresh review terms (pas due $\leq 30d$) • 10% Old review terms (past due $> 30d$) IF not enough terms to review, switch to SRS

▼ Full spec of term selection

SRS priorities

# Pri o	📍 Sour ce	Aa Name	≡ Description	≡ Data	☰ User is learning
1	Learn DB	<u>Available terms with next due date > 0 days</u>	Terms from Learn DB with timer not expired yet, starting with the timer that is the closest to expire.	Memorized = False LastStatus = True Timer is not done yet ex. if a card is due in 1 day, and another card is due in 3 days, start with the card that is due in 1 day	Favorites History List
1	Learn DB	<u>Available terms with next due date > 30 days late</u>		Memorized = False LastStatus = True Timer is done >	Favorites History List

# Pri o	📍 Sour ce	Aa Name	≡ Description	≡ Data	☰ User is learning
				30 days ago ex. if a card is late 47 days, start with it before doing a card that is late 31 days	
1		<u>New terms</u> <u>from DB Selected</u> <u>list</u>	We should start from most recently created.		Favorites History List
1		<u>Available</u> <u>terms I've</u> <u>studied</u> <u>recently</u>	Terms from the Learn DB, with an expired SRS timer ≤ 30 days ago - The priority is for the timer that expired the longest ago. - The logic for the timer is described below.	Memorized = False LastStatus = True Timer is done $\leq$ 30 days ago ex. if a card is late 3 days, start with it before doing a card that is late 1 day	Favorites History List
1		<u>Learn DB</u> <u>Wrong terms</u>	<i>Terms from the Learn DB, that I got wrong last time I practiced them.</i>	Memorized = False LastStatus = False	Favorites History List
2		<u>Discover</u> <u>Words to</u> <u>Discover /</u> <u>Suggestions</u>	Only if the user chose All Pick random terms from Word and Expression of the Day when available. These terms should not be in the user's Favorites or History. Here we should make an API call to add an		History

# Pri o	🕒 Sour ce	Aa Name	≡ Description	≡ Data	☰ User is learning
			example to the History term.		
2	History DB	<u>New terms from History</u>	Only if the user chose All or History Here we should make an API call to add an example to the History term.	<i>Here we should do multiple API calls to Context to retrieve 3 translations and 6 examples. The examples must have a source example of max 250 char.</i>	History
3	Learn DB	<u>Mastered terms from the list</u>		Memorized = True	Favorites History List
3	Favorites DB	<u>New reversed terms from List</u>	Let's say the user chose the direction en → fr IF the user doesn't have any terms in en → fr - Take fr → en terms and add them reversed in the Learn DB Don't reverse a term and is marked as ignored in Learn DB.	<u>For example:</u> If the user has "manger → eat" we add in the Learn DB "eat → manger". - The source becomes the target and the target becomes the source. - The source example becomes the target example, and vice versa.	Favorites History List
4	History DB	<u>Reversed terms from History</u>	Only if the user chose All or History Here we should make an API call to add an example to the History term.		History

# Priority	📍 Source	Aa Name	≡ Description	≡ Data	☰ User is learning
			Don't reverse a term and is marked as ignored in Learn DB.		

SRS Logic

SRS Timer

- If `LastStatus` is False, timer = 0
- If `LastStatus` is True, timer follows this table:

SRS Counter	Timer
0	
1	1 day
2	3 days
3	7 days
4	14 days
5	Mastered

Terms that shouldn't be added to Learn DB

- Source terms `<3` characters or `>24` characters
- History terms that are `Rude`
- History terms with `No results`
- History terms with Smart Suggestion → Take the Smart Suggestion only
- Type = Synonyms
- Type = MT

▼ Edge cases

Scenario	Behavior
< 7 terms available with timer done	Launch Flashcards with less than 7 terms. IF we have 1-6 remaining terms, return the remaining terms only. IF we have 0 remaining terms, the user should still be able to learn a list:

Scenario	Behavior
	Scenario 1 I launch a Quiz and all terms have SRS counter not ready: IF no timer is available, take the terms with the smallest remaining timer and let the user play. Scenario 2 I launch a Quiz and all terms are Mastered: reset SRS progress for Mastered to 0 and offer them again in the game, but keep the Memorized=True always. Scenario 3 I launch a Quiz and I don't have enough available terms: take in priority the SRS counter not ready, then the Mastered.
No more terms compatible with game	Return error message and display user pop-up.
If we have multiple terms with same Source and Target	IF the Source-Target already exists in the Learn DB, what do we currently do on mobile app? - Do not create a new entry in the Learn DB - Add a 2nd example to the examples array of the same entry IF the Source already exists in the Learn DB but with a different Target, what do we currently do in mobile app? - Add a new entry in Learn DB - Mark it as isExcludedfromQuiz = True so that it doesn't appear in the games

What to display on the card

Use case	What to display on card
Mastered = True	Mastered (with green check mark)
Nb view = 0	NEW (with purple icon)
Nb view ≠ 0 + Last Status = True	Progress X/5 (with pie chart)
Nb view ≠ 0 + Last Status = False	Failed last time (with orange repeat icon)

Updating Learn DB after a game

What happens at the end of a game?

User action	Update to Learn DB
Click on "Don't know" SRS counter = 0, 1, 2 or 3 isMemorized = False	Add +1 to TimesSeen Update LastSeen date Update LastStatus to False Remove -1 from SRScouter
Click on "Know" SRS counter = 0, 1, 2 or 3 isMemorized = False	Add +1 to TimesSeen Update LastSeen date Update LastStatus to True Add +1 to SRScouter
Click on "Don't Know" SRS counter = 4 isMemorized = False	Add +1 to TimesSeen Update LastSeen date Update LastStatus to False Remove -1 from SRScouter
Click on "Know" SRS counter = 4 isMemorized = False	Add +1 to TimesSeen Update LastSeen date Update LastStatus to True Update SRScouter to 0 Update isMemorized to True
isMemorized = True	Add +1 to TimesSeen Once the card is reset, same behavior as above BUT the isMemorized remains True no matter the user's answers (correct or false).

When are cards updated exactly?

Game	What is currently updated in this game?	What we want as new behavior	When is the card updated?
SRS Flashcard game	Everything in the table above / UPDATE: This game marks a card as memorized after a user swipes right or taps on the "Know" button 2 times on the last iteration. (compared with the	We want to see the card 5 times to mark as Mastered... so not 2 games, but 5 games.	At the end of the game, after 3 rounds only. During the game, nothing happens.

Game	What is currently updated in this game?	What we want as new behavior	When is the card updated?
	classic 5 stage SRS on the other games)		
Quiz - Multiple choice	Everything in the table above		Each time a user taps on a Quiz answer.
Quiz - Flashcard & Multiple choice	Everything in the table above		Each time a user taps on a Quiz answer.
Speak	Cards that get 2 correct answers across the phases are marked as memorized		1st correct answer → partially then the 2nd correct answer marks it as memorized
Autoplay	The game adds a "+1" in SRS each time a card is seen	Add +1 to TimesSeen <u>only</u> Update LastSeenDate and NbViews <u>Do not</u> change the SRScounter in this game	On each card swipe.
View all cards	Everything in the table above	Add +1 to TimesSeen Update LastSeenDate and NbViews Update LastStatus to True or False <u>Do not</u> change the SRScounter in this game	On each card swipe.
Word to discover	If a user taps on "Already Know: we mark the card as Memorized If a user taps on "Should learn" we add the card in Learning		when user taps on any of the buttons

Game	What is currently updated in this game?	What we want as new behavior	When is the card updated?
	the term is added to favorites		

▼ Example

Let's take this example to illustrate all the points above:

Initial state of databases

Favorites DB	Target	Creation date
punch		December 2
spike		November 3
laugh	rire	October 2
laugh	smile	July 1

History DB	Target	Creation date
split		December 1
hunch		November 2
dream		October 1
just		July 6

Learn DB	SRS Counter	Availability
apron	3/5	Available. Timer done.
colander	2/5	Available. Timer done.
whisk	✓ 5/5	Memorized. Not available anymore.

The user plays Flashcard

As you can see, we only have 2 available terms in the Learn DB. We first need to add 5 new terms to be able to get 7 in total that are available for learning.

Learn DB	SRS Counter
apron	3/5
colander	2/5

Learn DB	SRS Counter
whisk	✓ 5/5
punch	0/5
spike	0/5
laugh (rire)	0/5
split	0/5
hunch	0/5

Now the user can play Flashcards:



Flashcards

- apron ✓
- colander ✗
- punch ✓
- spike ✓
- laugh ✗
- split ✗
- hunch ✗

We update the Learn DB based on user performance

Learn DB	SRS Counter	Last Status	Timer
apron	4/5	✓ Correct	Timer = 14 days
colander	2/5	✗ Wrong	Timer = 0s
whisk	✓ 5/5	✓ Correct	Memorized for long term.
punch	1/5	✓ Correct	Timer = 1 day
spike	1/5	✓ Correct	Timer = 1 day
laugh (rire)	0/5	✗ Wrong	Timer = 0s
split	0/5	✗ Wrong	Timer = 0s
hunch	0/5	✗ Wrong	Timer = 0s

The user decides to play another session:

We have 4 terms with Timer = 0.

We need to add 3 more terms to Learn DB.

If we don't have any new terms, we can take the terms with the timer that is the closest to being finished.

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▼ Real-life scenarios

Study Scenarios

Aa Scenario	🕒 Type	≡ Current behavior	➤ SRS Solutions
<u>The user opens the app and spends all their time reviewing old cards. They never reach new content.</u>	Chill learner	This happens for users that study at their own pace, every now and then	<u>NEW strategy, MIXED strategy as default</u>
<u>I play one quiz per week</u>	Chill learner	I see the same 7 words for 3 weeks in a row	<u>21 cards before repeating</u>
<u>I have a list of 100 terms, I've mastered 30 and learning 70</u>	Medium lists	If due date is not reached for 70 terms, show Mastered again	<u>Never study MASTERED before finishing all terms</u>
<u>I study some terms, then stop studying for one month.</u>	After inactivity	After one month, I'll repeat the terms I studied one month ago, that I might not care about, before studying terms I've recently added.	<u>30 days rule</u>
<u>I have a big list of 1000 terms</u>	Big list	I review the same terms many times, it's very hard to review my whole list	<u>NEW strategy.</u>
<u>I want to review all cards before my exam tomorrow. I don't want to repeat.</u>	Exam	If you study everything the same day, you'll see everything. But if you study over 3 days... you'll repeat a lot.	<u>NEW strategy.</u>
<u>When I play 3 sets per session</u>	Routine learner	I'm always having the exact same sets of 7 (unless I make a mistake)	<u>SHUFFLE rule</u>

Additional rules

Aa Name	Description	Study Scenarios
<u>21 cards before repeating</u>	In Spaced repetition mode, do not start repeating terms with LastStatus=True until the user has 21 cards from the List in the Learn DB, this will avoid the user repeating the first 7 cards 3 times before seeing new ones (in the case the user plays once a week).	<u>I play one quiz per week</u>
<u>30 days rule</u>	When terms haven't been reviewed for a while, deprioritize them (just keep a small percentage in the game, rather than them being a majority)	<u>I study some terms, than stop studying for one month.</u>
<u>NEW strategy</u>	When the user navigates to "Not started" tab and launches a game, launch it on New terms only.	<u>I have a big list of 1000 terms, I want to review all cards before my exam tomorrow. I don't want to repeat., The user opens the app and spends all their time reviewing old cards. They never reach new content.</u>
<u>Never study MASTERED before finishing all terms</u>	If due date is not reached for 70 terms, show the term with closest next due date. Do not show any Mastered terms before Learning terms ever.	<u>I have a list of 100 terms, I've mastered 30 and learning 70</u>
<u>MIXED strategy as default</u>	When studying History or Big Favorites, have 40% new terms. When I launch a game, have 3 new terms and 4 terms based on SRS.	<u>The user opens the app and spends all their time reviewing old cards. They never reach new content.</u>
<u>SHUFFLE rule</u>	When picking cards to play in a Quiz, pick among all cards available for a given <u>day</u> and shuffle. Don't take seconds, minutes or hours into account. Or else, we end up with the same sets of 7 every time.	<u>When I play 3 sets per session</u>

